Module 07: Communication in Embodied

Learning Objectives

1. Define **Communication** within the context of Embodied.
2. Analyze how Communication interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Communication.

Introduction

This module explores **Communication**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Communication is a critical component of the 8-part Active Inference spine, bridging the gap between Learning and Planning.

Key Concepts

1. Communication as a Markov Blanket Boundary

How does Communication define the boundary between the agent and the environment?

2. Generative Models of Communication

What parameters involved in Communication must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Communication drive the perception-action loop?

Applications

In Embodied, we see Communication manifest in: * **Specific Example 1**: An experienced negotiator intuitively senses the moment to make a concession -- not from any explicit signal but from a subtle shift in the room's energy that she registers as a loosening in her own shoulders and a slight forward lean in her counterpart; this intuitive communication operates across social Markov blankets, where her generative model has learned to decode micro-postural and tonal cues into predictions about the other party's internal state, transmitted as embodied knowing rather than propositional analysis. * **Specific Example 2**: A mother intuitively distinguishes her infant's hungry cry from the pain cry from across the house, feeling a corresponding pull in her breasts for the first and a sharp alertness in her solar plexus for the second; this communication channel between parent and infant operates through deeply evolved embodied priors in the generative model, where the acoustic signal at the Markov blanket is instantly matched to a somatic prediction that simultaneously identifies the infant's state and initiates the appropriate caregiving response.

Conclusion

Understanding Communication allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.